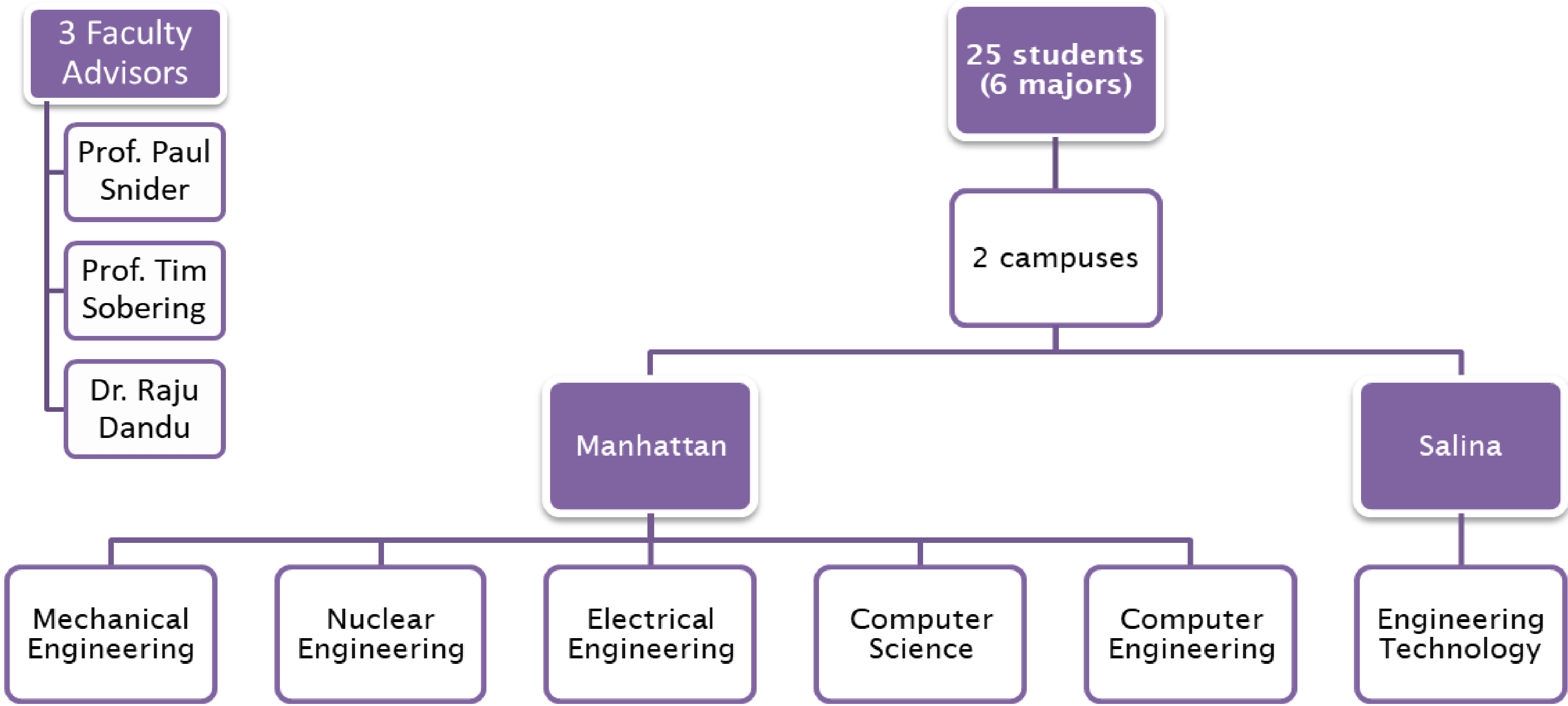




MANHATTAN-SALINA COLLABORATION



SCIENCE BACKGROUND

Purpose: Advance agricultural stability through high-resolution hyperspectral imaging

Problem: Current multispectral satellites and UAVs

- Lack of sufficient spectral detail
- Low revisit frequency (16 days) or infrequent data collection
- Cannot detect early crop stress or soil degradation

Agricultural Context:

- U.S. Midwest produces 40% of global corn, soybeans and wheat
- U.S. crop sector grossed \$275B in FY22, with 20% lost to avoidable issues
- 72% of farm profits spend on inputs, such as fertilizer and irrigation
- Resource management reduces overuse of water and chemicals, preserving aquifers and soil health

Impact:

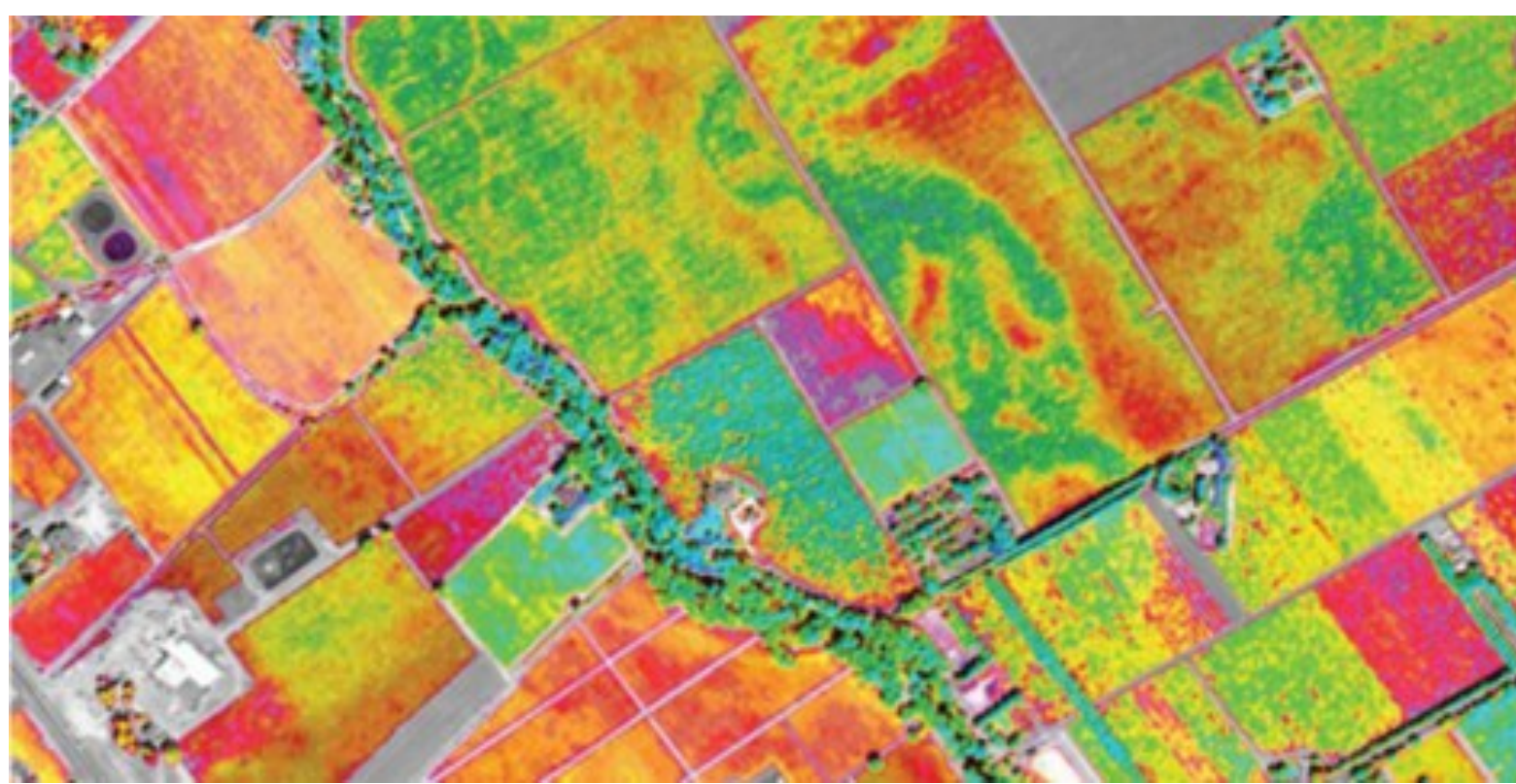
- Reduce economic losses
- Minimize environmental impact
- Empower Midwest farmers
- Supports K-State, NASA, USDA, and NSF goals

Index	Purpose	Formula
Normalized Difference Vegetation Index	Used to assess overall plant health and biomass	$NDVI = \frac{NIR - RED}{NIR + RED}$
Enhanced Vegetation Index	More accurate measurement of dense vegetation or high atmospheric disturbances	$EVI = G \cdot \frac{NIR - RED}{NIR + RED + 2.5}$
Normalized Difference Water Index	Measures moisture content in vegetation and soil to assess drought conditions	$NDWI = \frac{GREEN - NIR}{GREEN + NIR}$
Plant Senescence Reflectance Index	Used to detect early signs of plant senescence and stress	$PSRI = \frac{RED - NIR}{BLUE}$
Soil-Adjusted Vegetation Index	Reduces the effect of soil background in areas with low vegetation cover	$SAVI = \frac{(NIR - RED)(1 + L)}{NIR + RED + L}$
Leaf Area Index	Used to measure vegetation canopy structure and estimate plant productivity	Calculated with reflectance data model
Modified Soil-Adjusted Vegetation Index	Minimizes effect of soil in areas with low vegetation cover and provides better sensitivity	$MSAVI = \frac{2 \cdot NIR + 1 - \sqrt{(2 \cdot NIR + 1)^2 - 8 \cdot (NIR - RED)}}{2}$
Chlorophyll Index	Used to estimate chlorophyll concentration and overall plant health.	Combination of green and NIR or red-edge bands

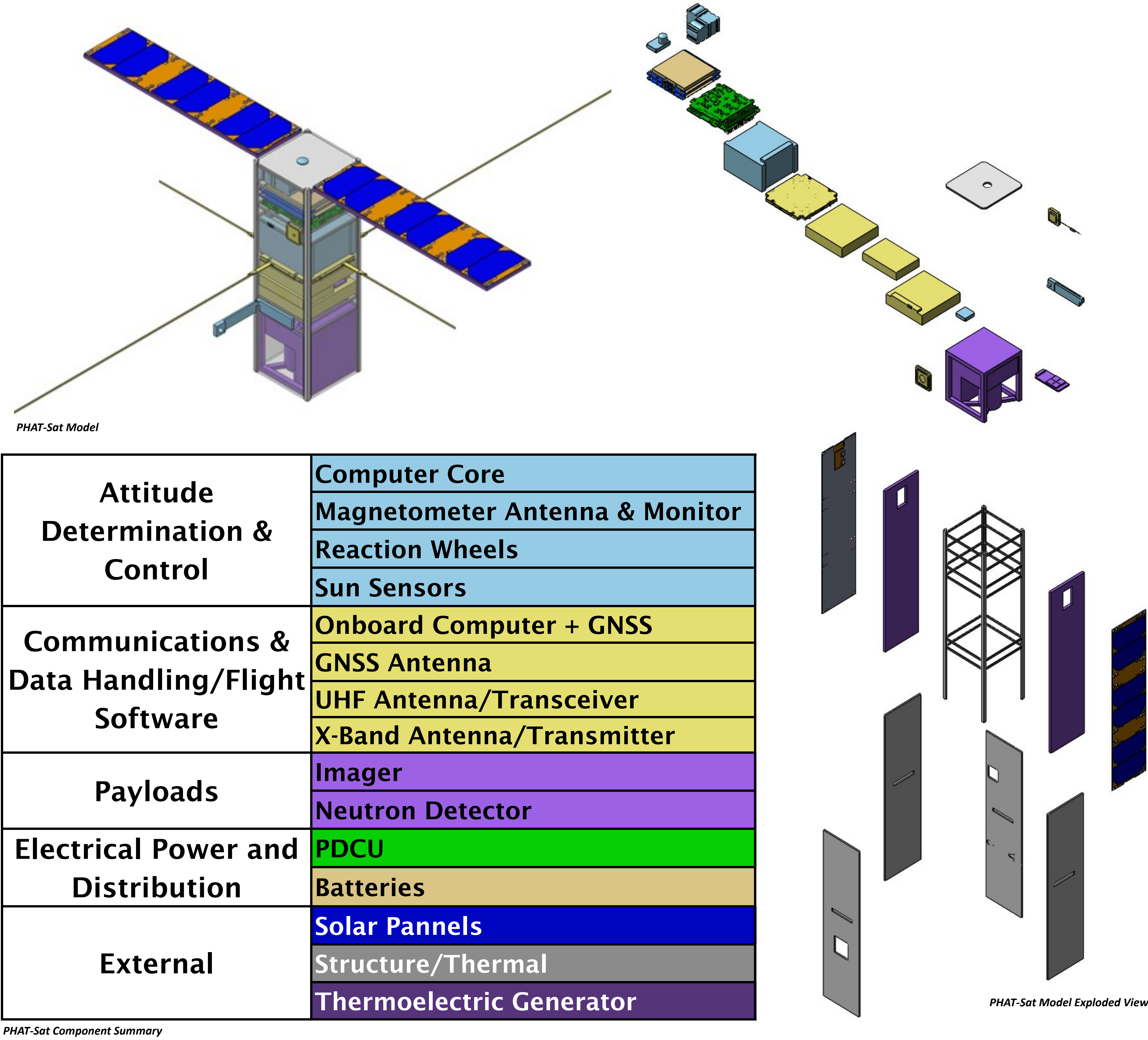
Common Vegetation Index Summary

VEGETATION INDICES

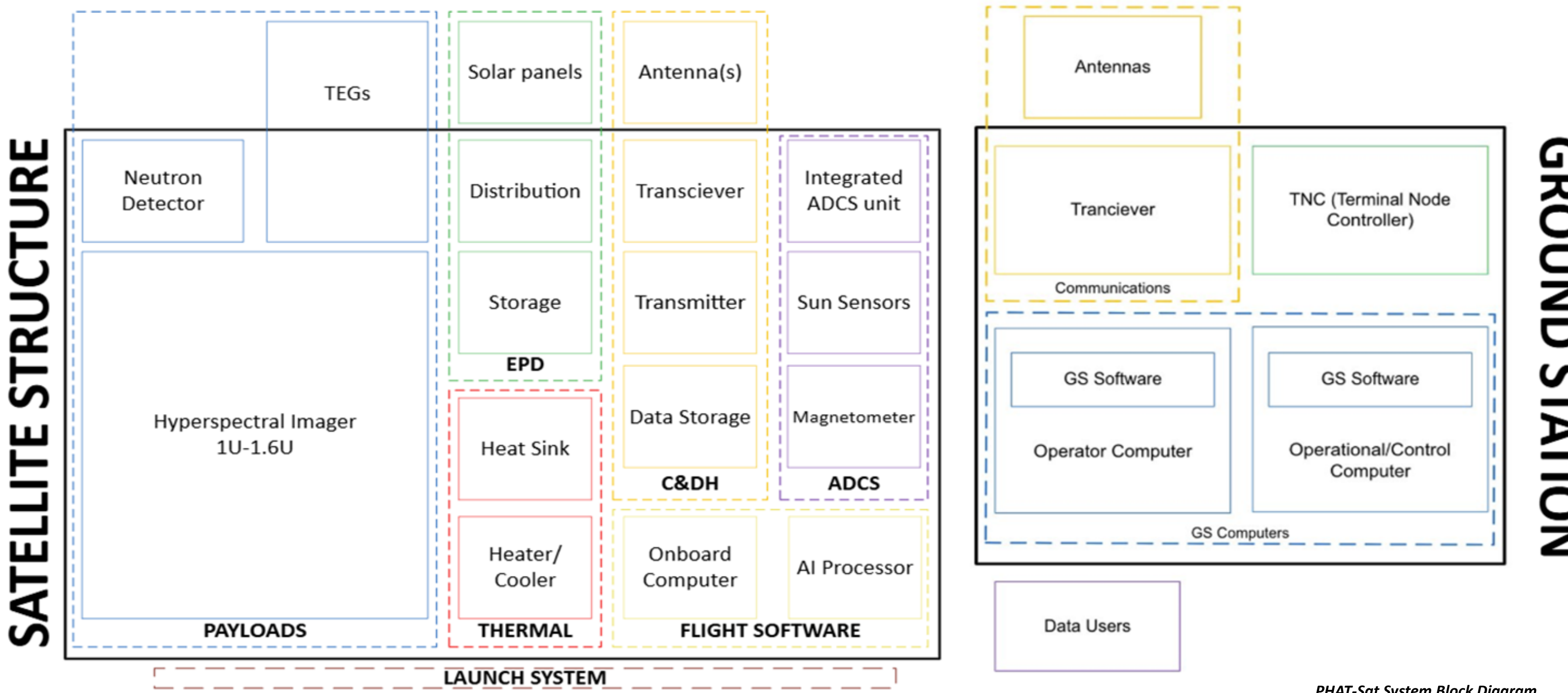
- Mathematical combination of reflectance values of certain spectral bands
- Used to quantify plant health, crop stress, water context, soil quality, etc.
- More spectral bands = Sharper Image



SATELLITE LAYOUT



PHAT-Sat Component Summary



PHAT-Sat System Block Diagram

MISSION SPECIFICATIONS

Size: 3000 cm³ (3U)

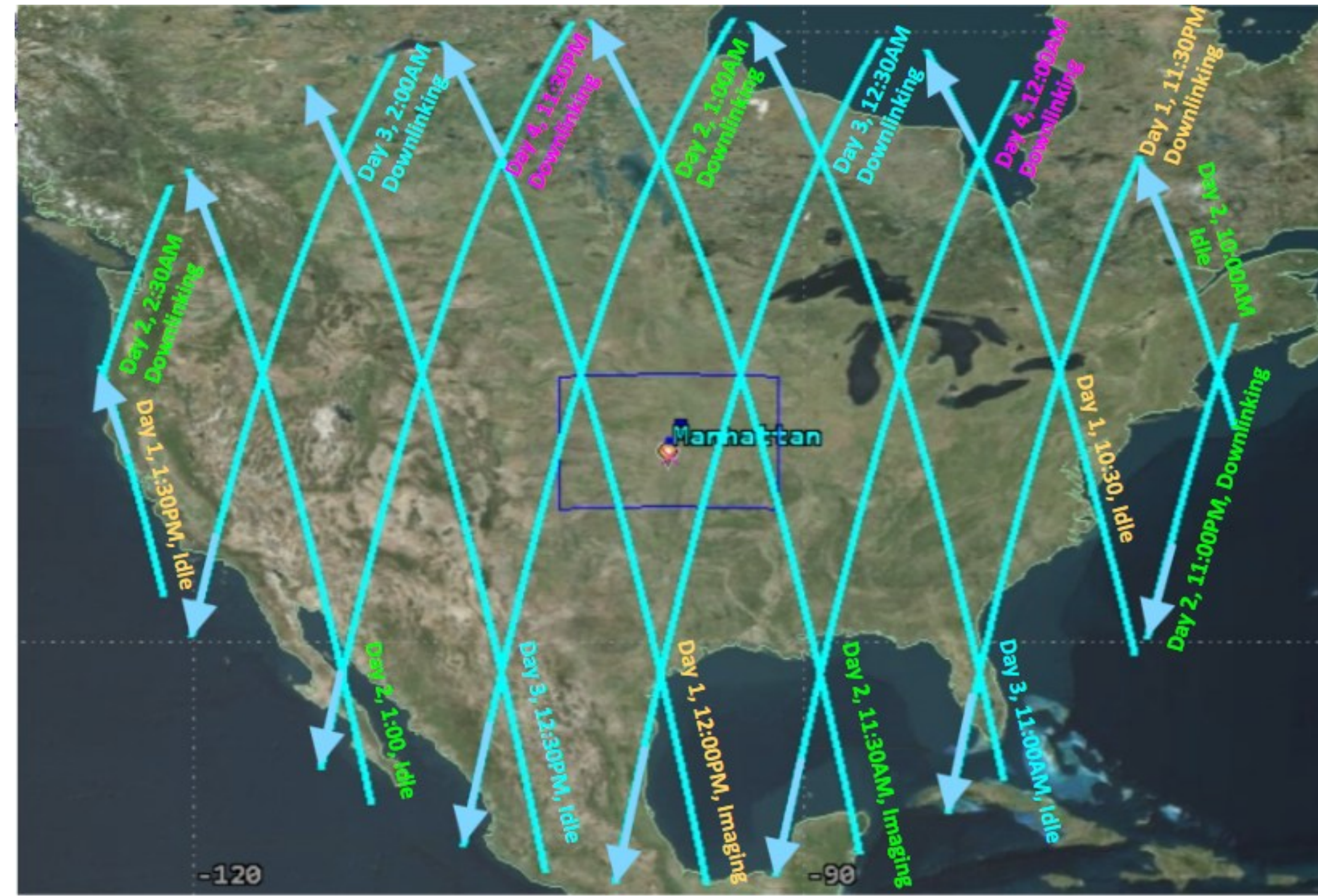
Payloads: Hyperspectral imager, miniaturized neutron detector, thermoelectric generator panels

Orbit: Sun-synchronous Low Earth Orbit (~500km)

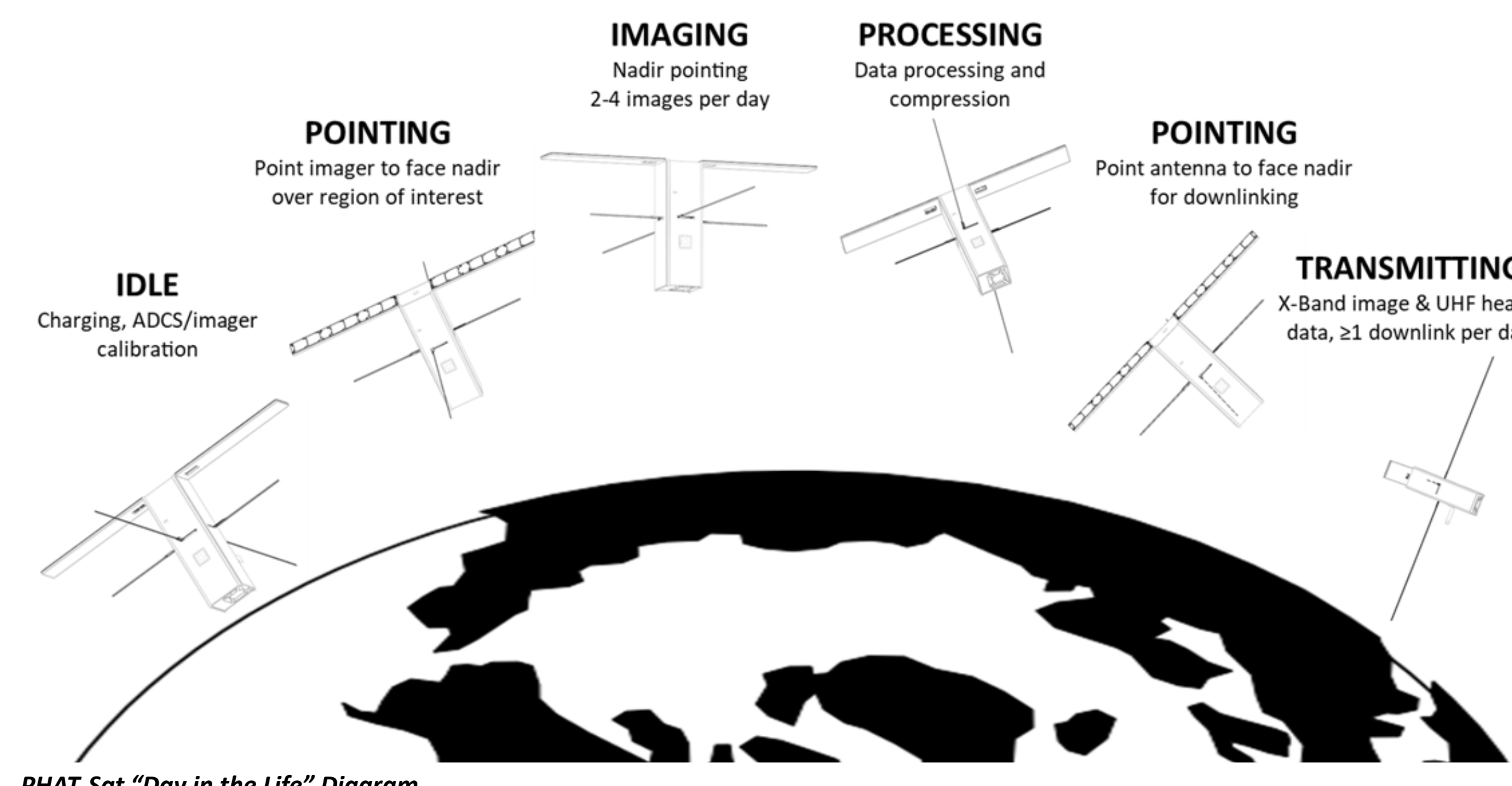
Mission Life: 24 months

Hyperspectral Imaging:

- >12 spectral bands
- Improvement of current imaging satellites
- 50m spatial resolution
- Supplementing current imaging satellites
- Regular coverage over key agricultural region
- Kansas, Nebraska, Iowa, Missouri, Oklahoma, eastern Colorado



PHAT-Sat Orbital Passes



PHAT-Sat "Day in the Life" Diagram

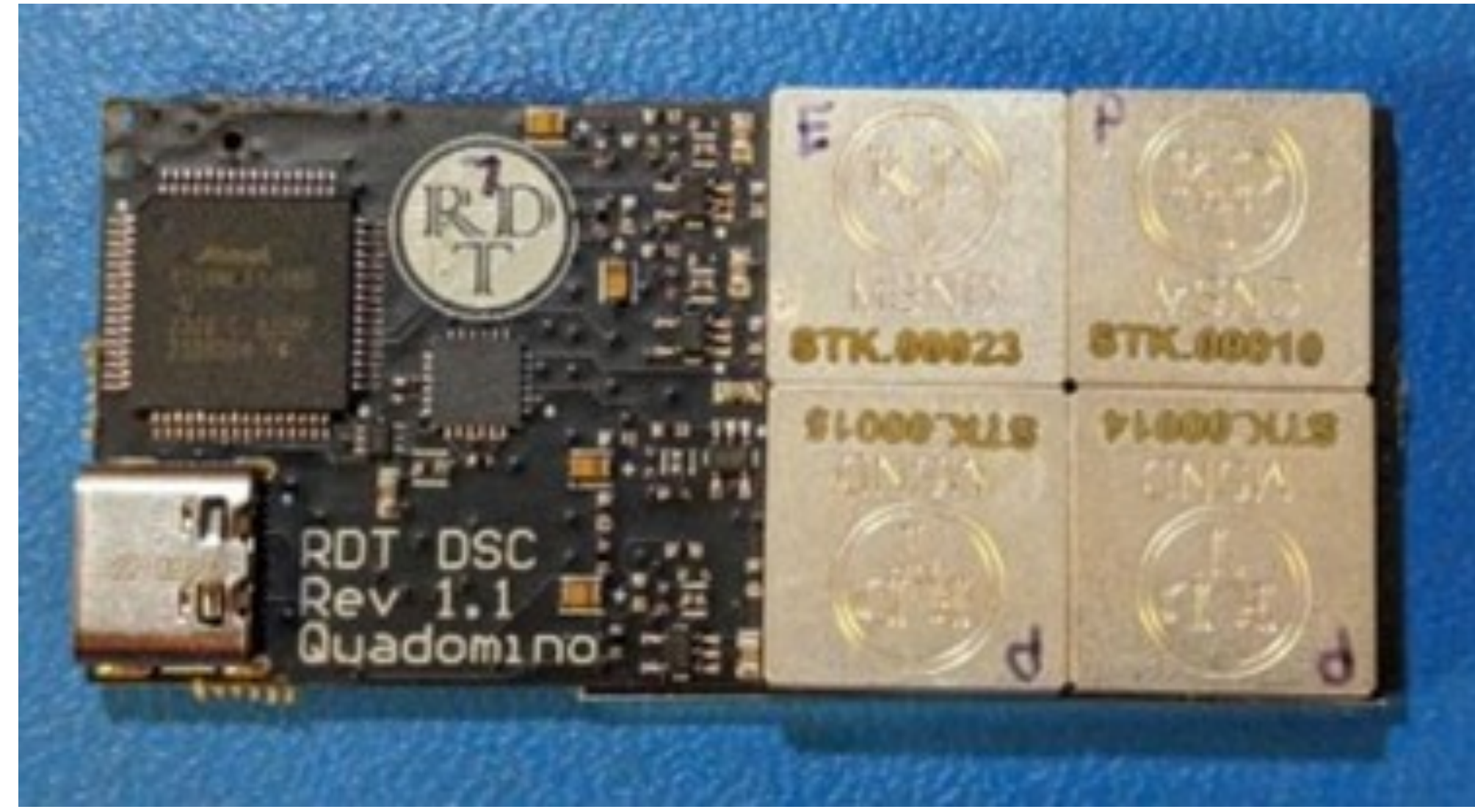
PHAT-Sat Sample Band Selections

CHANNEL	WAVELENGTH (um)
GREEN	450
BLUE	500
	550
RED	600
	670
	680
	800
	850
NIR (Near Infrared)	900
	1000
	1100
	1200
	1400
	1600
SWIR (Short Wave Infrared)	1700
	1800
	2000
	2200
	2400

HOSTED PAYLOADS

Neutron Detector Hosted Payload:

- DSC-Domino Designed and Built by K-State's RDT
- Miniaturized fast and thermal neutron detector (2-group measurements)
- Establishing detector space heritage and improving K-State's presence in space detection
- Microstructured Semiconductor Neutron Detector (MSND) to compete with bulky ³He detectors
- Energy and volume efficient with built-in background filtering
- Improvement on the Standard Domino currently on the ISS (strictly thermal neutrons)
- Improves TRL for future space missions



Neutron Detector, image courtesy of Radiation Detection Technology